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Interface Management in Gigafactory Design

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Gränssnittshantering vid design av Gigafabrik

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Sammanfattning

Stora batteriproduktionsanläggningar, ofta kallade Gigafabriker, är en komplex designuppgift som involverar många arbetsdiscipliner. Detta kräver goda förmågor för kommunikation och planering för att kunna integrera byggnaden, maskinerna och operatörerna i fabriken.

Detta examensarbete genomfördes i Factory Layout and Modelling-teamet i Blueprint på Northvolt, som är ansvariga för att designa och skala upp nästa generations batterifabriker. Syftet med examensarbetet är att öka kunskapen och vidareutveckla ett standardiserat arbetsflöde för att hantera maskingränssnitt.

De empiriska resultaten från studien baseras på en kvalitativ studie där 16 intervjuer genomfördes med personer som arbetar som fabriksdesigningenjörer, system-distributionsingenjörer, automationsingenjörer och projektledare. De empiriska resultaten bekräftar att det finns ett behov av ett standardiserat sätt att märka upp gränssnitt tidigt i designfasen för att minska risken för att designfel uppstår i senare skeden. Baserat på de empiriska resultaten är de viktigaste gränssnitten system-distributionsgränsnitten, materialflödesgränssnitten, människa-maskin-gränssnitten och byggnadsgränssnitten. Det är också viktigt att märka upp operatörens rörelse runt maskinen samt de områden där reparation och underhåll utförs. Detta skapar en förståelse för maskinens effektiva fotavtryck inom fabriken och visar den totala arbetsytan som behövs runt maskinen. I detta projekt har specifika designriktlinjer och en ny ritningsmall föreslagits för att kunna underlätta visualisering och hantering av maskingränssnitt.

Gränssnittshantering är ett framväxande ämne för produktionsföretag. Under det senaste decenniet har detta område utvecklats som en egen ledningsdisciplin inom vissa stora företag. Sammanfattningsvis är användningen av gränssnittshantering nödvändig för att öka kommunikationen av systemgränser och för att minska kostnaderna när nya fabriker byggs.

Nyckelord: Gränssnittshantering, fabriksdesign, byggnadsinformationsmodellering

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Abstract

Large battery production facilities, often referred to as Gigafactories, are a complex design task that involves a lot of working disciplines. This requires good capabilities for communication and planning to be able to integrate the building, machines, and operators in the factory.

This project is made within the Factory Layout and Modelling team in Blueprint at Northvolt, which is responsible for designing and scaling up the next generation of battery Gigafactories. The purpose of this thesis is to increase knowledge and further enhance a standardized workflow for how to manage machine interfaces.

The empirical findings from the study are based on a qualitative study where 16 interviews were held with people working as Factory Layout Engineers, Utility Engineers, Automation Engineers, and Design Leaders. The empirical findings confirm there is a need for having a standardized way of marking up interfaces early in the design phase to mitigate the risk of design errors to occur in later stages. Based on the empirical findings the most important interfaces are the utility interfaces, material flow interfaces, the human-machine interfaces, and the civil, structural, and architectural interfaces. It is also important to mark up the operator's movement around the machine and the areas where repair and maintenance is needed. This creates an understanding of the machine's effective footprint area and shows the total workspace area needed around the machine. In this project, specific design guidelines and a new drawing template have been proposed to facilitate the visualization and handling of machine interfaces.

Interface management is an emerging topic for production companies. During the last decade, this area has been developed as a management discipline within some large companies. In conclusion, the usage of interface management is necessary to increase communication of system boundaries and to decrease the cost when building new factories.

Keywords: Interface management, Factory Design, Building Information Modelling

FOREWORD

This master thesis was written in spring 2024 at the Royal Institute of Technology in Stockholm, and the study was made at the battery manufacturing company Northvolt.

First, I would like to thank my industrial supervisor Axel Save for the opportunity to investigate this topic and for supporting me during this project. Additionally, thanks to the Factory Layout and Modelling team at Northvolt and all the respondents who provided useful information on how the interface management workflow could be further developed.

I would also like to thank my academic supervisor Jenny Janhager Stier for your guidance, support, and insights which have been very helpful during this project.

Gabriel Dalborg

Stockholm, June 2024

NOMENCLATURE

This section covers the abbreviations used throughout the report.

Abbreviations

AEC	Architectural, Engineering, and Construction
AGV	Autonomous Guided Vehicle
APS	Autodesk Platform Service
BIM	Building Information Model
CSA	Civil, Structural and Architectural
EV	Electric Vehicle
GTS	General Technical Specification
HMI	Human-Machine Interface
IFM	Integrated Factory Model
IM	Interface Management
LOD	Level of Detail. A construction methodology to define development stages
MEP	Mechanical, Electrical, and Plumbing
MVP	Minimum Viable Product. A basic product used for testing and feedback.
PO	Purchase Order
RFQ	Request for Quotation. A process before purchasing a product or service.

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SAMMANFATTNING (SWEDISH)

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1 INTRODUCTION

In the introduction chapter, the background, purpose, and delimitations are presented.

1.1 Background

Due to the regulatory shift towards sustainability and electrification, the global demand for batteries increases exponentially every year and is estimated to grow from 700 GWh in 2022 to 4700 GWh in 2030 (McKinsey & Company, 2023). The largest battery demand comes from the mobility sector and the transition towards electric vehicles (EVs). Most companies within this sector are now phasing out the development of combustion engines and fully focusing on electric solutions. From a global perspective, 14 % of all new vehicles were electric in 2022, compared to 9 % in 2021 and 5 % in 2020 (IEA, 2023). In 2030, electrical vehicles are estimated to account for 40 % of all global sales (Fornaro & Hummel, 2021).

The European Commission established in 2023 a net-zero industry act to exceed the manufacturing capacity of clean energy technologies in the EU. The proposal from the European Union's Net Zero Industry Act is that European manufacturers should be capable of covering 90 % of the EU's demand for batteries in 2030 (European Parliament, 2023). Europe is the second largest sales market of EVs, where over 20 % of all newly registered vehicles were electric in 2022 (IEA, 2023). In Sweden, battery production is becoming a key industry that creates thousands of new jobs and strengthens the Swedish business community (Swedish Energy Agency, 2023). The ongoing battery development will play a key role in the global transition to renewable energy and electrical vehicles (Crownhart, 2023). The Swedish battery manufacturing company Northvolt, established in 2016, has a goal to produce the world's greenest battery cell. The long-term goal within the battery industry is to create a fully circular economy through recycling and recovering raw materials (BIS Research, 2024). Northvolt's mission is that 50% of all raw material for cell production will come from recycled batteries in 2030 (Northvolt, 2024).

The transition towards sustainable battery production demands green and smart factories to be built. The Architectural, Engineering, and Construction industry (AEC) is responsible for designing, planning, and building infrastructure and factories. This industry is evolving at a fast pace through new technologies that optimize overall performance and management (Biswas et al., 2023). Building Information Modelling (BIM) is a best-practice process and 3D modelling software which contains all the detailed information regarding the building project (Green, 2016). The introduction of BIM has been considered a new paradigm within the AEC industry since it increases collaboration between all stakeholders (Azhar, 2011).

When designing a factory, there are many design disciplines involved working on different system implementations. Between these systems, there are system interfaces that need to be managed effectively throughout the process and that is a challenge today. If these are not acknowledged and integrated into the system, there is a risk for serious interface issues to occur in later stages (Zander et al., 2022). Due to the increased complexity of construction projects nowadays, Interface management (IM) is emerging as an own management discipline (Shokri et al., 2015).

This study is made within the Blueprint Factory Layout and Modelling team at Northvolt, which is responsible for designing, developing, and scaling up the next generation of battery Gigafactories. A Gigafactory is a large-scale facility that produces batteries, and the name relates to the gigawatt hour (GWh) production capacity these factories are capable of (Whiteaker, 2022). The Blueprint team is working in a constant iteration process for updating and developing the factory production planning through an Integrated Factory Model, and the team now wants to further develop the interface management workflow to enhance coordination and communication for all projects.

1.2 Purpose

The purpose of this project is to increase knowledge about interface management and support a standardized factory design workflow for how to manage machine interfaces when designing new Gigafactories. All machines in the factory have different interfaces, for example, the physical attachment to the floor, utilities coming in and out, or human-machine interfaces. The thesis will present an investigation of how to develop the Integrated Factory Modelling workflow so these interfaces can be marked up in the model. A large focus of the project will be on how the 3D model can be optimized to enhance communication in the initial phase of a new project and between all design disciplines. The thesis will also analyze the opportunities, challenges, and risks of implementing a new workflow. As part of the thesis work, recommendations on future improvements will be written, e.g. CAD guideline updates and new standards which could be used.

1.3 Delimitations

This project has been delimited to only investigate machine interfaces, such as material flow interfaces, utility interfaces, human-machine interfaces, repair and maintenance interfaces, and the interfaces between the machine and the civil, structural, and architectural elements. Other types of elements, that do not have a direct connection with the machine will not be investigated.

2 FRAME OF REFERENCE

This chapter goes through the different levels of factory layouts and how the models are developed and integrated into the overall Building Information Model. The chapter also goes through how interface management is acknowledged in the industry today.

2.1 Factory Layout and Modelling

A factory layout can be seen as an allocation plan where structural units are spatially positioned (Wiendahl et al., 2015). The layout can be outlined at different levels, ranging from macro representation showing a complete factory site overview to micro representation showing the detailed positioning of machinery and resources (Wiendahl et al., 2015). These different types of layouts can be called the site layout, rough layout, detailed layout, and workstation layout as shown in Figure 1. A site layout shows the building and road placement and is also referred to as the master building plan (Wiendahl et al., 2015). The rough layout shows the individual production areas within the factory and the logistics of material flow routes within the factory. Within the detailed layout or fine layout, the positioning of equipment and the energy and media supply is shown. The layout can also be further detailed to show the workstation layout which outlines the micro representation for a precise location of machinery, tools, energy supply, and media supply (Wiendahl et al., 2015).

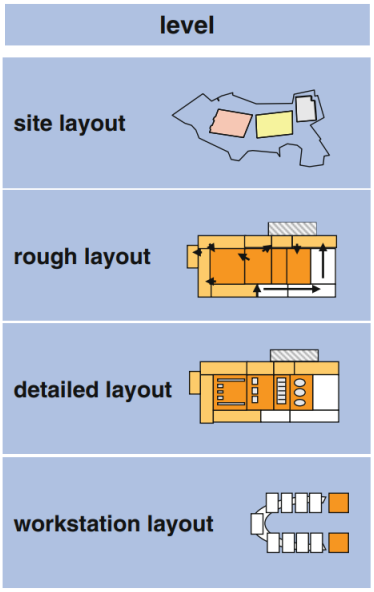


Figure 1. Types of Layouts (Wiendahl et al., 2015)

A well-designed factory layout increases efficiency and productivity and decreases the total costs of building the factories (Collins, 2023). To accomplish this, several aspects need to be considered while designing a new factory layout. The flow of materials, personnel, and products should be considered to minimize the transportation distance between the production workstations (Collins, 2023). Space utilization is important to minimize the factory footprint and overall cost. Besides this, the safety aspect of employees and equipment is important to mitigate the risk of errors and accidents to occur (Collins, 2023). Due to rapidly changing business environments, it is important

to consider flexibility according to the study by Weckenborg et. al. (2023). The principle of Lean Manufacturing should also be considered to eliminate waste in the form of material, labor, and time but also to increase speed, efficiency, and quality (Rouse, 2015).

2D layouts are later converted to 3D layouts showing ceiling clearance (Wiendahl et al., 2015). Digital 3D factory models are used for both visualization and creation of advanced computational designs (Sharma, 2023). The introduction of Building Information Modelling (BIM) is considered an important step in the AEC industry history due to its capability to reduce time, and cost and at the same time increase productivity and quality (Azhar, 2011). This new technology has nowadays outcompeted the traditional methods of orthogonal drawings and specifications (Abrishami et al., 2020). A Building Information Model (BIM) is an intelligent digital representation of a construction project that enhances the coordination through the project's entire lifecycle (Autodesk, 2024). BIM increases productivity by enabling early design optimization, clash detection, and schedule integration which also reduces the project's total costs (Biswas et al., 2023). The term "Integrated Factory Model" is also used when talking about large production projects. According to Autodesk (2024), *"Integrated factory modelling combines BIM and digital factory planning to generate a digital representation of production facilities and equipment".* These advanced digital tools are constantly updated with new features which enhance the overall performance of construction projects.

Throughout an in-depth expert knowledge study by Burggräf et. al. (2019), three main obstacles within factory design were observed. The first obstacle is correlated to the cost and complexity of designing the MEP(H) since it requires strong collaboration and planning. The second obstacle is the unaligned level of detail and lack of data transmission between the different design disciplines. The third obstacle observed is the constantly changing requirements and lack of understanding of the associated change effects according to Burggräf et. al. (2019). The production lines within the factory are in constant change and development, especially due to technological advancements. The changes can be driven by new product launches, cost reduction initiatives, local regulations, standardization changes, or changes within the procurement and supply chain. The demand for adaptable factory systems is emerging, and this sets high demands on universality, scalability, modularity, and compatibility in the factory, according to Horbach et al. (2010).

There is an industry BIM standard for Level of Development (LOD). The LOD standard allows AEC professionals to communicate and specify deliverables based on six design stages: LOD 100, LOD 200, LOD 300, LOD 350, LOD 400, and LOD 500 (BIM Forum, 2023). In LOD 100, machines are modelled as 2D symbols. In LOD 200, machines are modelled with approximate size and location. In LOD 300, the models have more precise geometry and in LOD 350 the models also show the connections and interfaces with other model elements. LOD 400 and LOD 500 then show more detailed representations of the elements (BIM Forum, 2023).

2.2 Interface Management

A factory can be seen as a complex product that follows the theory of technical systems (Francalanza et al., 2017). Morris et al. (1988) early stated the importance of Interface management and having a system perspective approach for large, complex, and urgent projects. He mentioned

that systems are logically broken down into subsystems, which create system boundaries and interfaces (Morris, 1988). The control of interfaces is crucial for program and project success (NASA, 2023). The general workflow is to define the interfaces, identify the documentation and drawings for complete configuration, and set up a process to ensure interface compatibility (NASA, 2023). Pavitt & Gibb (2003) categorized interface management into three main categories: physical, contractual, and organizational interfaces. The physical interface refers to the connection between two or more building elements, while the contractual interfaces refer to physical interfaces created when elements are grouped as a work package (Pavitt & Gibb, 2003). The organizational interfaces refer to the interaction of various teams that are involved in the construction process.

Shen et al. (2021) created a framework for Interface management performance based on formal governance, partnering, and bounding activities. Within a regression study, formal governance and partnering have a strong correlation on Interface management performance and overall project outcomes (Shen, et al., 2021). Partnering is considered a significantly important factor in large-scale projects, and according to Shen et al. (2021) "measures should be taken to constantly establish and enhance trust, openness, and communication between stakeholders for achieving project goals in terms of schedule, cost and quality". Overall, the study by Shen et al. (2021) shows that the enhancement of Interface management leads to a more effective and efficient flow of information for companies.

Due to the increased complexity of construction projects, some companies have separate departments dealing with Interface management (Shokri et al., 2015). Based on the study by Shokri et al. (2015), most of the projects that utilize Interface management are megaprojects with values of about one billion dollars. The result from the study shows that a systematic use of interface management can reduce overall cost growth (Shokri et al., 2015). Managers in construction are mainly focusing on cost, time, and quality but these objectives will not be reached unless interface management is considered (Keerthanaa & Shanmugapriya, 2017). The benefits of using interface management are for example better understanding of project complexity, reduced interfacial issues and uncertainty, and updated coordination between participants (Chen et al., 2007). The result of not using interface management could be design errors, mismatching parts, and system failure.

Chen et al. (2008) further examined the importance of having a system approach for interface management to mitigate and resolve interface issues. Through a multi-perspective analysis, Chen et al. (2008) identified six broad areas that cause interface issues within organizations; these areas are based on people/participants, methods/processes, resources, documentation, project management, or the environment. The first approach, people/participants, could be caused by unawareness of interfaces, ownership, and responsibilities or by lack of coordination and communication in the project. The second approach, Methods/Processes, could be caused by construction, manufacturing, or assembly issues. The third approach, Resources could be caused due to lack of labor or equipment, while the fourth approach, Documentation, could be about lack of interface documentation, database storage, or inadequate drawings. The Project Management approach could be a result of poor interface management or inappropriate subcontracting while the Environmental approach could be caused by local regulations or trade union practices.

Insufficient communication of Interface changes is a large issue within Factory planning. Cheng (2012) addressed this issue by creating a web-based ConBIM-IM platform where participants and stakeholders can manage the changes. According to Chen et al. (2007) Interface management is also important in Lean construction for understanding complexity and in Agile project management for achieving efficient teams. The integration of Interface management into BIM is a way of improving project monitoring and enabling real-time decision-making (Hmidah, et al., 2022). The utilization of algorithms and automation in BIM is considered very important, but there are still issues to resolve before high performance can be achieved (Hmidah, et al., 2022). Wiendahl et al. (2015) said: *"The problem with algorithms to optimize layouts is that almost all of them pursue only one optimization criterion – usually: minimal material flow or greatest space utilization…"* and further said, *"…this is far too simple, especially when it comes to fostering soft targets like changeability and communication."* Abrishami et al. (2020) investigated automation for concept design optimization by utilizing BIM and generative design. According to Abrishami et al. (2020), there are future challenges to these implementations since the end-users often lack programming experience and do not know how to write and develop the algorithms. Further on, Yarmohammadi et al. (2018) presented a way to track performance and modelling data and how it can be collected in real time directly from the BIM model. These are some examples of automation systems that can improve factory design, communication, and management.

2.3 Machine Interfaces

Wiendahl et. al. (2015) mentioned the importance of understanding the total machine workspace area, which consists of the machine base area and the space required for material supply, maintenance, waste disposal, and the operator movement around the specific machine as shown in Figure 2. Further on, Wiendahl et. al. (2015) suggested that the machine information, including views, dimensions, connections, and data can be displayed on an ID card to effectively balance the communication between the space planners and the production engineering.

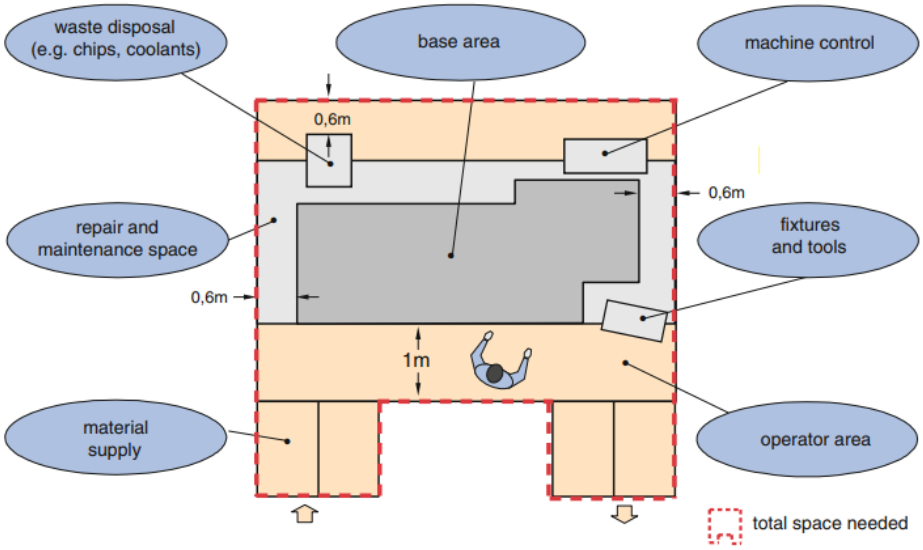


Figure 2. Total machine workspace area based on the surrounding area components. (Wiendahl, Reichardt, & Nyhuis, 2015).

According to Wiendahl et al. (2015), it is highly necessary to document all these requirements early on and provide them in a way so the information can be updated easily during the process. When the spatial planning of machinery and equipment is started, a lot of different interfaces need to be considered. It is also crucial to consider these interfaces when factories are restructured (Hellmuth et al., 2020). The supply and removal of energy and media is one important interface. This refers to a physical utility interface between the machine and the process elements. Other types of physical interfaces are the material flow transportation between the workstations in the factory, or the interfaces between the machine and the civil, structural, and architectural elements such as the physical attachment to the floor. Besides this, there will also be a flow of operators throughout the factory, and the operators will be interacting with the machines. Figure 3 shows some of the most important interfaces to consider when starting a new factory layout.

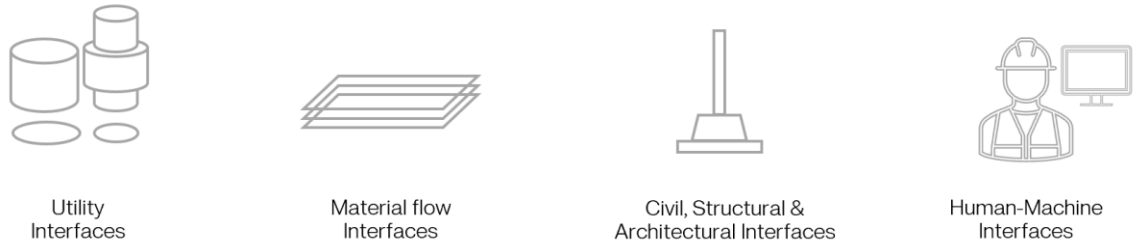


Figure 3. Physical Machine Interfaces to consider in early design stages

2.4 Research questions

Parallel development of machines and layouts, in combination with rapid technological advancements, is common in the battery industry. This requires construction management practices, such as interface management, to overcome complexity and improve alignment and communication (Shokri et al., 2015). This study aims to investigate which changes and improvements could be made within the Factory layout and modelling workflow to improve interface management. Based on this, two research questions have been defined:

RQ1: What are the main challenges within factory modelling and interface management today?
RQ2: How can these interfaces be managed more effectively when designing Gigafactories?

3 METHOD

The method chapter outlines the study design and presents more in detail how the literature study and the qualitative study were carried out.

3.1 Study design

This thesis is based on a literature study and a qualitative interview study. Before the interview study, a pre-study was conducted to get a broader understanding of how interfaces are handled today at the company. In total, 13 people were interviewed in the pre-study and the interview study. Three of these people were interviewed twice since they were considered extra important due to their current knowledge and experience within the field. The interviews were transcribed and then clustered to form requirements and recommendations for how the interface management workflow could be developed. Figure 4 shows an overview of the process steps in the study.

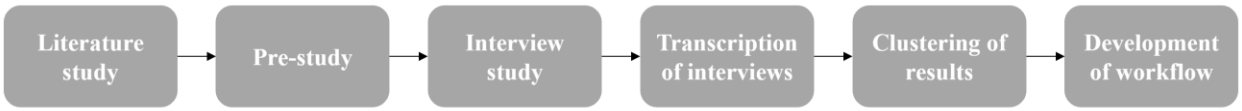


Figure 4. Process steps from start to finish in this study.

3.2 Literature study

A literature study was carried out as a first step to get a wider understanding of factory design issues and how interface management and system thinking can be used to handle this. The literature was published in journals within engineering, construction, architectural management, and sustainability. Some of the sources also come from books and websites within Factory planning and interface management. To find these books, journals, and articles, the academic databases Scopus and Web of Science were primarily used. The following keywords have been used to find literature:

- Factory planning OR Factory design OR Factory modelling OR Facility planning
- Interface management OR Interfaces OR Machine connections
- BIM development OR BIM automation OR Factory layout automation

3.3 Pre-study

In the pre-study, interviews were conducted with 8 people at the company and these interviews took between 20-30 minutes to complete. These interviews were not recorded, although notes were taken during the interviews. All respondents within the pre-study have a technical background in engineering and have many years of experience in designing Gigafactories and large-scale production facilities. A description of the respondents and their roles at the company is shown in Table 1.

Table 1. Respondents in the pre-study discussions

Role	Description	Date	Place	Length
Developer	Driving development of modelling workflow and software updates	1/9/2024	Online	20 min
Layout Engineer 1	Creating conceptual process layouts	1/11/2024	Office	30 min
Information Specialist	Years of experience in BIM and building Gigafactories	1/12/2024	Office	30 min
Utilities Engineer 1	Responsible for building systems that generate, store, and distribute process utilities	1/12/2024	Office	30 min
Automation Engineer 1	Working with automation systems in the factory	1/15/2024	Office	20 min
Layout Engineer 2	Creating detailed process layouts	1/18/2024	Office	30 min
Utilities Engineer 2	Calculating utility and energy consumption for the factory	1/18/2024	Online	30 min
Design Leader 1	Experienced in Construction management of Gigafactories	1/22/2024	Office	30 min

3.4 Interview study

In the interview study, 8 interviews were conducted. These interviews were based on semi-structured interview guidelines, see Appendix A and Appendix B. All interviews took between 20-35 minutes to complete and all the interviews from the second iteration of interviews were recorded and later transcribed through an automated transcription tool called KTH play. The transcripts were later reviewed, edited, and clustered into different themes in Excel based on the type of interface. Besides these interviews, there has been ongoing weekly discussion with the Factory Layout and Modelling Manager, who has been the mentor for this thesis project. The respondents within the interview study are shown in Table 2.

Table 2. Respondents in the interview study

Role	Description	Date	Place	Length
Simulation Engineer	Responsible for simulating the factory production lines and analyzing the design	2/6/2024	Online	35 min
Design Leader 2	Driving the design of controlled room environments around the process	2/8/2024	Online	30 min
Utilities Engineer 1	Responsible for building systems that generate, store, and distribute process utilities	2/9/2024	Online	30 min
Layout Engineer 3	Creating detailed process layouts	2/9/2024	Online	35 min
Information Specialist	Years of experience in BIM and building Gigafactories	2/15/2024	Office	20 min
Utilities Engineer 2	Calculating utility and energy consumption for the factory	2/15/2024	Office	30 min
Design Leader 3	Responsible for integrating all disciplines to create the factory design	2/16/2024	Online	30 min
Automation Engineer 2	Working with material flow handling and automation systems for transportation	2/26/2024	Office	30 min

3.5 Development of workflow

A systematic and iterative approach for investigating, developing, and testing was used to identify new ways of working and to optimize how interfaces can be marked up in the factory modelling environment. The recommendations and workflow development have been ongoing for 8 weeks, and the factory layout and modelling team have participated in two feedback sessions where they have guided and given recommendations for the newly developed solutions.

3.6 Validity of study

The selected people for the qualitative study have a lot of experience from different projects within Northvolt and have specific insights into the current process. The interview study was based on semi-structured interview guidelines to offer guidance and for structurally asking the questions. However, there has been deviations from the guideline in some interviews since the respondents are specialized within one specific area and could not answer all questions. One delimitation of the study is that only internal interviews has been conducted. This means that no external consultants, experts, or equipment suppliers have been interviewed. Although no external people have been interviewed, the results provide a good understanding of the internal ways of working and which areas that need further investigation.

4 EMPIRICAL FINDINGS

The empirical findings chapter presents the results from the interviews. Due to confidentiality reasons, some of the results are not presented here.

4.1 Interview study

Within this chapter, the most important findings from the interview study are presented. The findings have been clustered based on different types of interfaces: Utility Interfaces, Material flow interfaces, CSA interfaces, and Human Machine interfaces. The chapter ends with explaining standards and additional findings from the study.

4.1.1 Utility Interfaces

Utility Interfaces are considered the most important interface based on the interview answers. To be able to plan for pipelines, it is necessary to have an initial overview of where the interfaces will be located according to the respondents. The utilities can be broken down into utilities coming in (supply) and utilities coming out (return) from the machine. The respondents see potential in labelling interfaces earlier to mitigate the risk of misunderstanding between the different design disciplines. One of the Design Leaders mentioned the importance of ensuring resources for machine design and integration to make the process more efficient. By visualizing utility interfaces earlier, better communication with the engineering contractors can be achieved. However, the maturity level for the interface placement depends on the maturity level of the machine design. Although marking up interfaces is important, there are some concerns since detailed utility information can only be provided when the design reviews of the machine are finalized. As one of the Automation Engineers said:

"Based on my experience so far, interfaces keep on moving until you have a design review done and until you put a sign on a paper saying this is the final design." – Automation Engineer 2

At Northvolt, they design their factories within three separate phases: Concept design, Basic design, and Detailed design according to the respondents. In concept design, Northvolt creates different types of theoretical diagrams for the processes. These documents give a good overview of the full system and the utility distribution. When the factory layout is initiated in concept design, it is necessary to visualize the utilities needed for specific machines to enable pipeline planning, based on the answers from the respondents. In basic design, the layout and BIM will reach a higher maturity level. At this stage, the machine placement will be more accurate, and the quantity, position, and flow rate for each utility and machine are needed. At the end of Basic design, it is necessary to show the placement of utility interfaces within one meter accuracy according to one of the Utility Engineers. The pipe dimension for each interface is not necessary in Basic design. In detailed design, high-maturity interface information including pipe dimensions and the flow rates are needed.

Further on, it is also important to understand the number of interfaces. For example, if two machines are standing close to each other, they can share the same utility hook-up point. A utility hook-up is a place where specific utilities will be distributed. According to one of the Utility

Engineers, a full system understanding is needed before hook-ups can be placed in the models. Before all machine designs are reviewed and finalized, the placement of hook-up points can change. To create a full understanding of the factory and its functionality, the theoretical flow document, the layout, and the BIM model need to be combined. One of the Design Leaders said that marking up utilities is just a matter of having someone do it. However, a lot of different people need to be involved in this simple task, while there are many other things with high priority. Receiving all the required information from the equipment suppliers at the right time will also be crucial according to the respondents.

4.1.2 Material Flow Interfaces

According to one of the Automation Engineers, who is working on designing the material transportation within the factory, it is also important to display where the material for each machine comes in and out. This information needs to be both within the 2D process layouts and the 3D BIM model according to the automation engineer. This is especially important from an educational perspective when external people need to understand the material flow in the factory.

There are several types of autonomous guided vehicles (AGVs) within the factory, and it will be crucial to occupy space where these AGVs will drop and pick up material. If an AGV interface is not marked up, other designers may place sensors or other components where this is not possible, which causes clashes with the AGVs. The AGVs used for material flow transportation also have safety fields on the sides, which need to be displayed in the models to mitigate the risk of elements coming near during movement. The height of the AGV during its movement from point A to point B also needs to be visualized. These material flow interfaces need to be displayed early in the process, and the placement needs to be refined on the way.

The Automation Engineer mentioned the importance of having a BIM model that represents everything that will exist on the real factory floor, so all stakeholders have a common picture and do not try to occupy space where it is not possible. To create material flow simulations, it is important with material flow data, such as input and output speed according to the Simulation Engineer. The idea is to develop a bi-directional flow between the layout and the simulation, where both can feed each other useful information.

4.1.3 Civil, Structural and Architectural Interfaces

The Civil, Structural, and Architectural (CSA) interfaces refer to the connection between the machine and the building, such as the physical attachment to the floor and the structural building elements. The placement of pillars or other structural building components will be dependent on where the machines are placed in the factory. To mitigate the risk of these interface issues occurring, it is important to occupy space or block space where you know something will be placed later. During the design process, there is an ongoing feedback and optimization process of fitting the building around the process equipment, as the Design Leader said:

"It [the factory design process] is sort of like a feedback loop. We get a first iteration of a process layout, then I try to fit the building around it. I see opportunities for optimization, and I talk to the process layout team, so they make some adjustments. Then I optimize the building again and come back with some more optimization." – Design Leader 3

The physical height of the machine, plus the void space above the machine, is needed for all machines represented in BIM, according to one of the Design Leaders. The void space is needed for piping installation, accessing a hatch, or any kind of maintenance work. This is also the constraint for deciding how high the floor or roof above is going to be. Besides this, it would be good to show floor markings in the 3D model, according to one of the Automation Engineers. The sprinklers used for extinguishing a fire within the factory also have minimum clearance distances to the machines. which could be integrated into BIM. These sprinklers do not have direct contact with the machines in the factory, but the placement of these sprinklers is dependent on the machine's placement.

The Simulation Engineer further said it is important to critically question the tolerances set for the specific machines, such as the weight, vibration, and flatness criteria of the floors since these parameters set the price of the factory. The maturity level of this information then needs to be refined on the way when the layout and BIM are updated throughout the design phases.

4.1.4 Human-Machine Interfaces

Human-Machine Interfaces (HMI) refer to the interaction between the operators working in the factory and the machines. Based on the interview answers, it is crucial to reserve space for the operator's working area around the machine. It is also crucial to reserve extra space for repair and maintenance areas where, for example, cabinets, hatches, and doors are located. At Northvolt, the space needed around the machine is called the effective footprint. This information is needed in concept design so the right volume of space can be occupied in the process layout and in BIM. Receiving this information in basic design or detailed design is considered as too late since it is hard to expand those areas later, according to one of the Layout Engineers. All this information comes from the equipment suppliers, which need to create drawings where all of this is specified. To access the machines, there needs to be a people path heading to the machine.

Most machines in the factory also have HMI computer screens, which are used for controlling and monitoring the machine during production. According to the Simulation Engineer, the HMI control panels can both be mounted directly on the machine or hanging from a protruding arm from the machine to enable ergonomics for the operator. These different types of HMI panels need to be visualized clearly within the models.

4.1.5 Standards

One of the Layout Engineers explained how Northvolt receives information from the equipment suppliers. According to the Layout Engineer, Northvolt has a General Technical specification (GTS) and CAD Guidelines which comprise useful information and standards for factory models and machines. There is also a standard that comprises what and when specific information from the equipment suppliers is needed. These guidelines and standards may need to be revised and updated to receive the necessary interface information from the equipment suppliers. As one of the Layout Engineers said:

"It should be clear from the beginning what we ask for, and what we ask for is what should be delivered by the equipment suppliers. What we do not control in our standards, we leave open for the suppliers to deliver as they want." – Layout Engineer 3, translated from Swedish.

Further on, Northvolt utilizes the BIM ISO-19650 industry standard for Level of Development (LOD) which specifies the BIM design requirements. According to this standard, there are six levels of development that can be used to specify deliveries with suppliers and contractors.

4.1.6 Additional findings

Based on the answers from the respondents, it is important to have a full system understanding of the factory before the factory layouts and models are started. According to one of the Utilities Engineers, the starting point of a new project is to look at historical plants and have an inventory of model data that can be reused for the next project. After this, the layout and modelling of the factory can start. Many respondents mentioned the importance of reserving space for separate components that will come later in the model, such as utility equipment, cabinets, or doors. Reserving these space blockers will be important since a lot of different people are involved in the design process of the Gigafactory.

5 ANALYSIS

Within the Analysis chapter, the empirical findings are analyzed in comparison with the theoretical findings. The chapter also investigates which areas and tools that need to be further developed at the company to strengthen the interface management workflow.

5.1 Interfaces in a factory

Collins (2023) said a well-designed factory increases efficiency while decreasing the total cost of the factory. The introduction of new digital tools, such as BIM and Integrated Factory Modelling have changed the way factories are built today. It is now easier than ever to optimize the factories based on space utilization and transportation distances of material and personnel. As both the empirical findings and the literature study revealed, there are still obstacles within the factory design process that need to be addressed. These obstacles often occur when two separate systems meet, which creates interface issues that can only be solved by working cross-functional with different design disciplines. Francalanza et. al. (2017) said a factory can be seen as a large and complex product that follows the theory of technical systems. Having a system perspective approach for large and complex projects to handle subsystems and interfaces has been discussed for decades, for example by Morris et al. (1988). Newer studies, such as Chen et al. (2007) and Shokri et al. (2015), showed that a systematic use of interface management reduces overall cost, reduces uncertainty, and contributes to better coordination between participants.

The respondents in this study said that the supply and return of utilities, which is included in the MEP scope, is the most complex and important interface to consider while building a factory. The largest issue is that machines can be under development at the same time as the factory layout is outlined. The position and capacity of utility data cannot be fully determined before the machine design is mature enough. The markup of interfaces is thereby driven by the progression of the machine design. The expert knowledge study by Burggräf et al. (2019) also indicated that the cost and complexity of MEP is one of the main obstacles in factory design. Burggräf et al. (2019) further explained that unaligned levels of detail and constantly changing requirements are two other main obstacles.

Utility data and interface data need to be refined and updated during the way the factory is designed, hence it is important with a smooth changing procedure. According to the respondents, the minimum requirement for all machine models would be to indicate the type of utilities the machine needs. In the next stage, the type and capacity for each utility need to be estimated by the equipment suppliers and integrated into the factory model. The final stage would be to have highly mature utility data for all machines and hook-up points visualized in the factory. According to the BIM ISO-19650 standard for Level of Development, interfaces and connection with other elements should be shown in LOD 350. As described in the interviews, an earlier overview of this information is necessary to be able to plan for where the piping system should be routed.

At Northvolt, the total workspace area is called the effective footprint area. Based on previous experience in factory planning, the layout engineers mentioned the operator's workspace area and the maintenance area as especially important to show from the beginning. When initially reserving space around the machine, it is important to consider that the machine dimensions could change according to the respondents. Thereby, the maturity level of all interfaces will be low until the machine design is finalized. However, giving a first indication of these interfaces would help to increase coordination and collaboration based on the interview study. As described by Wiendahl et al. (2015), all machines have a base area and a total workspace area. The base area is the machine footprint in the factory, while the total workspace area also includes the areas needed for controlling and operating the machine, such as the operator's area, the material supply areas, and the repair and maintenance areas. Wiendahl et. al (2015) enhanced the importance of an early understanding of the total workspace area.

Standards and guidelines will be necessary to succeed with interface management according to the respondents in this study. Wiendahl et al. (2015) mentioned that machine information can be displayed on an ID card. The ID card can work as a standardized way of collecting machine information from equipment suppliers. The ID card can then be updated by the equipment suppliers during the machine design progression, from the initial drawing request to the final machine design review. The equipment suppliers will then be responsible for marking up the interfaces, but the factory designer will be responsible for converting and integrating this information into BIM. For all interfaces, there needs to be a change process in place, which describes how and when data should be updated.

On one hand, the quality of the inputs from equipment vendors plays a critical role according to the respondents. Without the right information about machines, interfaces, and data from the equipment suppliers, it is very hard to visualize the machines in the factories. On the other hand, the inputs from the equipment suppliers also need to be effectively converted and implemented into the overall factory design. This requires information to be updated and integrated easily during the design. Shokri et. al (2015) showed that some companies have separate departments dealing with interface management since it is such a complex area. Most of these companies are working on megaprojects with values of about one billion dollars. This indicates that the usage of interface management is dependent on the project size and complexity. Reaching the construction planning goals of cost, time, and quality will be heavily dependent on interface management, according to the study by Keerthanaa et al. (2017).

The process of integrating interface information can be automated to some extent; however, it requires metadata to be early integrated into the model. It also requires the data to follow a standardized format and to be consistent within the model. The automation can be done in different ways depending on the used program. Most CAD software has in-built automation functionalities which can be utilized. Before high-performance objectives can be achieved, the algorithms and automation need to be in place within the programs. Abrishami et al. (2020) mentioned there are challenges with this since the end users may not always have the expertise or time to create and implement this.

The integration of more interface information in BIM would increase coordination according to the respondents. One respondent also mentioned the importance of ensuring resources for machine design and integration to make the process more efficient. The study by Chen et al. (2008) also clarified that companies require capabilities and resources to succeed with interface management. Hmidah et. al. (2022) said the integration of interface management into BIM has many advantages, such as increased monitoring and decision-making. Through a regression study, Shen et al. (2021) showed how formal governance and partnering have a strong correlation to interface management performance and project outcomes. Shen et al. (2021) also mentioned the importance of taking measurements to establish trust, openness, and communication. When implementing an interface management workflow, it would be good to measure the number of design changes and cluster them based on the type of interface.

5.2 Summarized recommendations from data collection

Based on the data collected from the literature study and the interview study the following topics has been identified as important in interface management:

- A factory should be seen as a large and complex product that follows the theory of system thinking to handle subsystems and interfaces.

- An early understanding of the total workspace area around the machines is necessary to minimize the risk of design errors. The total workspace area includes, for example, operator areas, repair and maintenance areas, utility areas, and material flow areas.

- Standards and guidelines should include information regarding interfaces to enable effective communication between the equipment suppliers and the factory layout engineers.

- Interface data should be updated at each design stage. The maturity level of the interface data can be low, medium, or high depending on the design stage and if the machine design is finalized.

- Utilizing automation tools in BIM is crucial to make an effective interface management workflow.

6 DEVELOPMENT OF WORKFLOW

The following chapter describes the steps for how the interface management workflow can be updated based on the empirical and theoretical findings. The findings presented within this chapter need to be assessed for implementation. Due to confidentiality reasons, the models in this chapter do not show real machines and factories.

6.1 Standards

As part of the thesis project, recommendations have been given for how the standards could be developed based on the interface management aspect.

6.1.1 CAD guideline update

Northvolt's current CAD guidelines provide information about the 2D and 3D models and pinpoint in detail which views and information should be delivered by the equipment suppliers for each design phase. The overall goal of these standards is to ensure a smooth information flow between Northvolt and the external suppliers. One part of the thesis project has been to look at the potential to further enhance information sharing regarding interfaces within these guidelines.

6.1.2 Standard template update

As Wiendahl et al. (2015) suggested, a drawing template, also called ID card, can be used to collect machine information where views, dimensions, and data are visualized. An example of a drawing template that could be used is shown in Figure 5. The template is based on the isometric drawing standard view in first angle projection and includes interface elements to enable mark-up of utilities, material flows, people paths, and the repair and maintenance areas, which is used to define the total machine workspace area. The drawing template also shows where the machine operator will be standing when handling the HMI control panel. To mark up the mechanical piping for the utilities for the machines, the ISO 20560 color safety standard can be used. This standard is mainly used for safety purposes when physically marking up utilities in the factory. All interface elements and pre-defined utilities are listed on the right side of the drawing template.

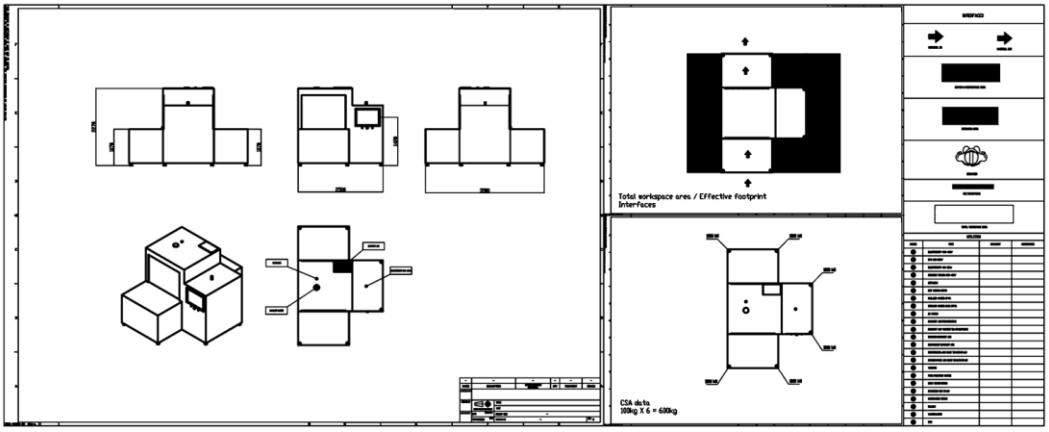


Figure 5. Example of drawing template

6.2 Interface development

As described in the interviews, there needs to be a standardized process for when and how different interfaces and utilities are shown in the BIM model based on the different design phases. When starting a new factory design layout, the maturity level of dimensions, interfaces, and data will be low as described in the interviews. This might be because the specific equipment has not been chosen, the machines are under development, or the design reviews and purchase orders (PO) have not been placed with the equipment suppliers.

6.2.1 Machine interfaces and total workspace area

The first step when designing a new battery Gigafactory will be to prepare theoretical documentations for the factory. When the factory layout starts, the largest element within the factory needs to be modeled in low-level detail, which will create a rough layout, showing the overall process steps and material flows. When the more detailed layout starts, the machines for the separate process steps will be placed out. At this level of detail, elements are visualized as generic placeholders with approximate quantities, sizes, and shapes in the 3D software according to the standard. As both the academia and the respondents embarked, it is important that the effective footprint of the machine, in other words the total workspace area, is marked out in the model. Without this information, there is a risk that the space reservation for each machine will be underestimated or overestimated.

Northvolt utilizes a wide range of software as part of their factory design toolbox, especially based in the Autodesk ecosystem. To build upon the company's existing toolbox, these programs have been used for development and testing. Figure 6 shows an overview of how a machine placeholder could look like in LOD 200. The model does not show any detailed information regarding the machine. This model shows the total workspace area around the machine, including the operator's interface with the machine (purple), the people's path (green), and the maintenance area (blue). The model also shows the material flow input (green arrow) and the material flow output (red arrow).

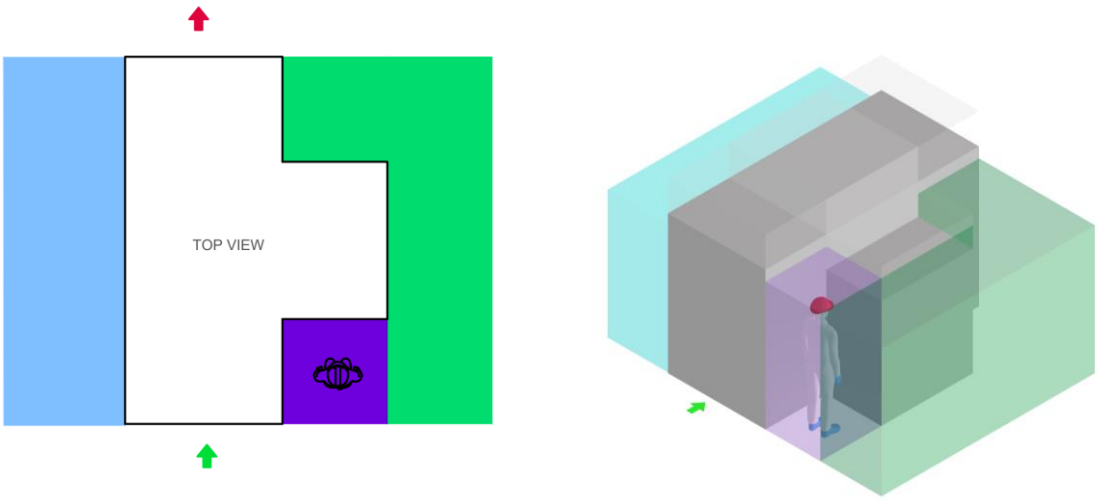


Figure 6. Example of machine within LOD 200. Left: 2D model. Right: 3D model.

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In the next design stage, the factory models will reach higher maturity levels, and the factory can be modelled in more detail on the machine level. This means the current placeholder block for the machine can be replaced with a more detailed 3D model showing, for example, the HMI display and the floor connection. The model will then reach LOD 300, and the size, shape, and location will be more accurate. An example of a machine modelled at LOD 300 is outlined in Figure 7.

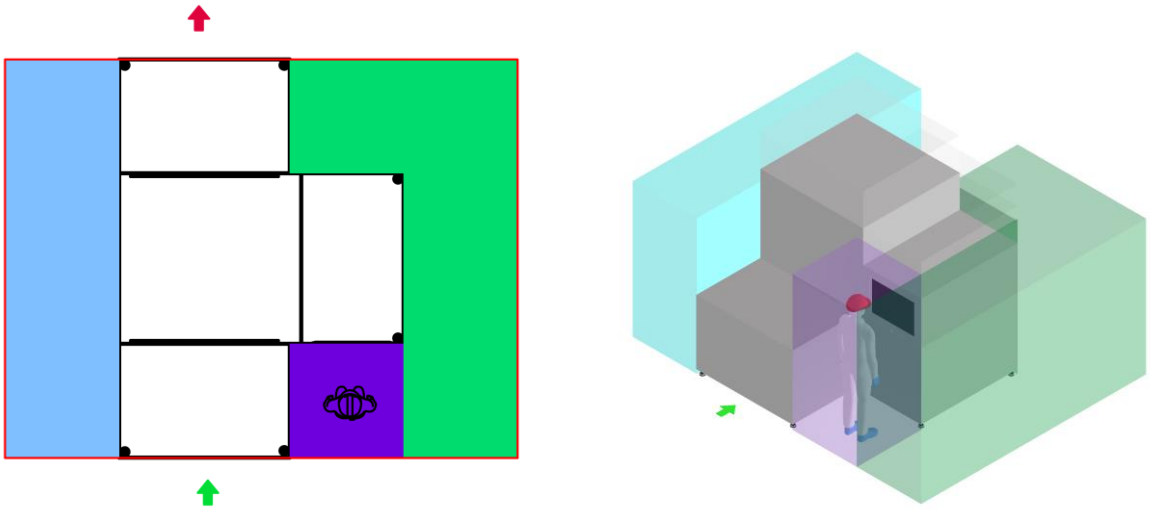


Figure 7. Example of machine within LOD 300. Left: 2D model. Right: 3D model.

As shown in Figure 7, no utility connections are marked up. The utility connections are normally not shown in the model until a request for quotation (RFQ) is made and the equipment suppliers send detailed information regarding the interfaces. When RFQ is accomplished, all the interfaces and utility points can be marked up in the model and the machine will then reach LOD 350. The metadata for each of the machines will also be updated with higher accuracy. This can be performed by combining the proposed drawing template with the 3D model. Figure 8 shows an example of a machine where all utility interfaces are marked up in LOD 350.

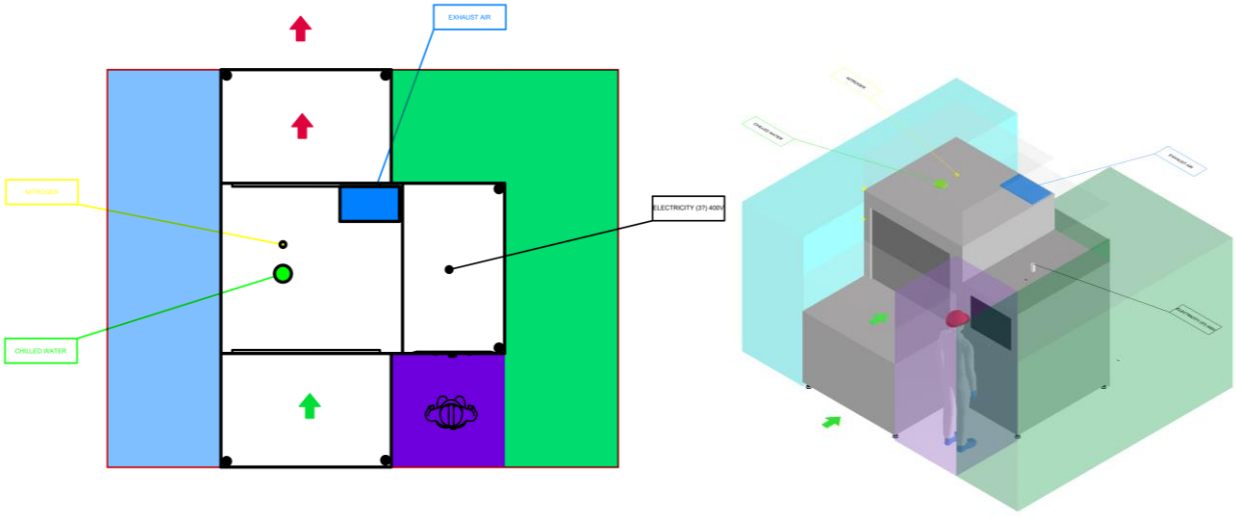


Figure 8. Example of machine within LOD 350. Left: 2D model. Right: 3D model.

By combining the detailed 3D model, with interfacial elements, a detailed representation of the machine is created. At this stage, it is also possible to attach MEP connectors and tags to each of the utility interfaces. From here, the machine has all the necessary components to be ready for Mechanical, Electrical, and Plumbing (MEP) design. Between each of these design stages, the production lines containing all the machines will be exported to the overall building information model (BIM) of the factory. The overall factory can be seen as a complex product, which includes three main systems; the building design including the architectural elements, the production planning including all machines, and the technical building equipment needed for distributing utilities and running the factory.

6.2.2 Interface markup in BIM

According to the LOD standard, interfaces and connections should be visualized in LOD 350. As described in the interviews, there needs to be a way how interfaces can be visualized earlier. The respondents said it is necessary to know which areas, processes, and units will consume what type of utility in the early stages of BIM design. Autodesk's platform enables different automation tools, such as iLogic in Inventor or Dynamo in Revit. The Building Information Modeling is currently created in Revit. Thereby, Dynamo was used to create a Minimum Viable Product to demonstrate how interfaces can be marked up. Dynamo is a graphical programming interface within Revit where custom workflows can be developed. Figure 9 shows the Dynamo program and Dynamo Player Interface used to perform this action. The same results can also be generated by creating view templates in Revit, where separate views can have predefined rules for controlling the visibility and override of the elements. These view templates can then be loaded into existing factory design projects.

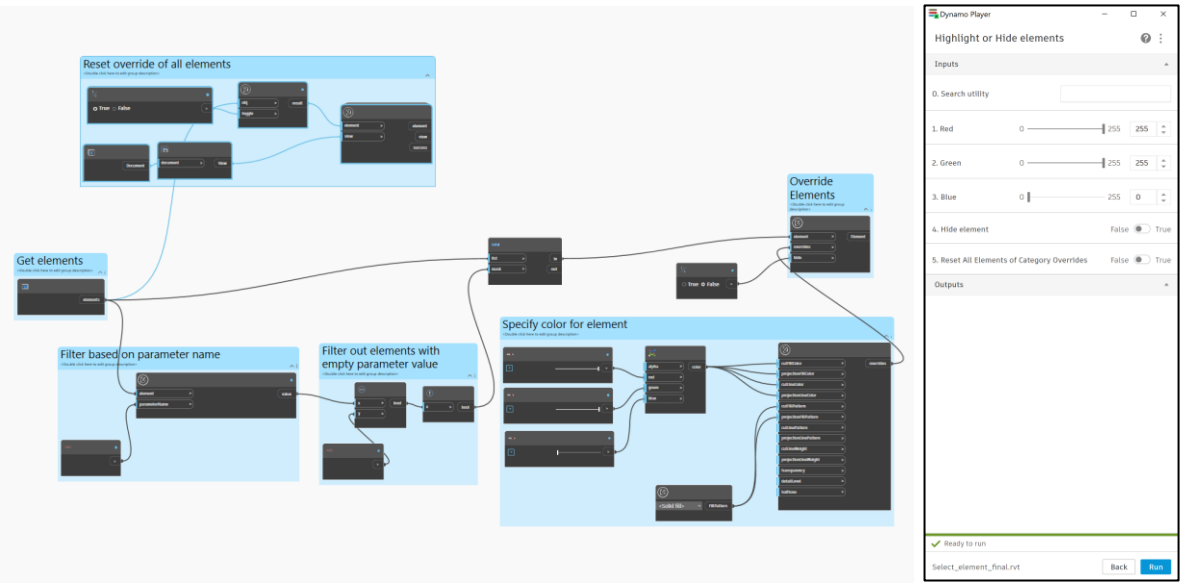


Figure 9. Overview of Dynamo program and Dynamo Player

This Dynamo program does not transfer data to the model elements. Therefore, it is required that each machine model element in the factory already have metadata attached. When all elements have metadata, the program can search through the elements within the model. Parameters without value will be filtered out. When searching for a parameter name in the Dynamo Player, all the corresponding elements containing that utility data will be highlighted on a separate drawing view.

The user can specify which color the interfaces should be highlighted in. By utilizing this Dynamo program, it will be possible to easier detect which areas, processes, and machines consume what utilities. This tool can thereby help with MEP coordination and hook-up placements. To test the program, an example machine layout has been created in Revit, see Figure 10. The layout contains machine elements, material flow arrows and people flow paths. The machine elements have generally been modelled as placeholder blocks for visualization purposes. To automate the flow of creating separate views of a model, all model elements must have separate data parameters. For example, to create a view where all people paths are hidden, the people path elements need to have a parameter that can be used as a reference. Therefore, all machine elements have been assigned parameters. Due to confidentiality, the test layout created in this section does not correspond to existing factory design projects.

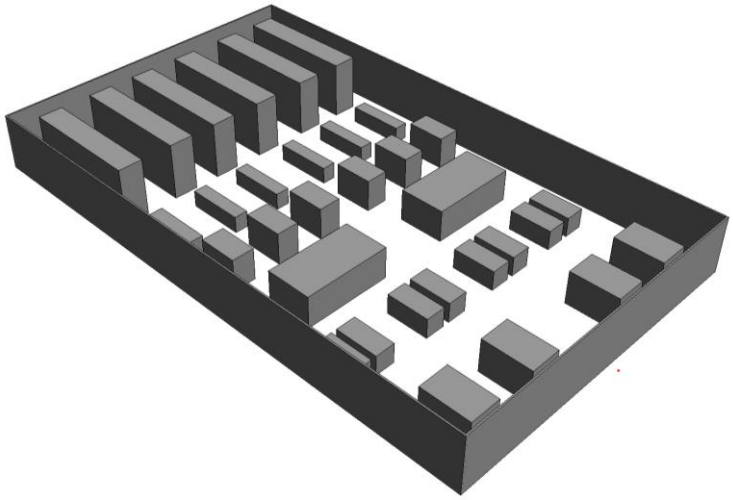


Figure 10. Test layout with machine placeholders

For the machine elements, it is possible to highlight or hide machines with specific properties. An example of highlighted machines based on utility consumption is shown in Figure 11. In this view, the entire machine element is highlighted. This is because no utility interface elements exist yet in the layout. In later stages, each machine can be assigned MEP connectors, and the position of these connectors can then be highlighted in a new separate view.

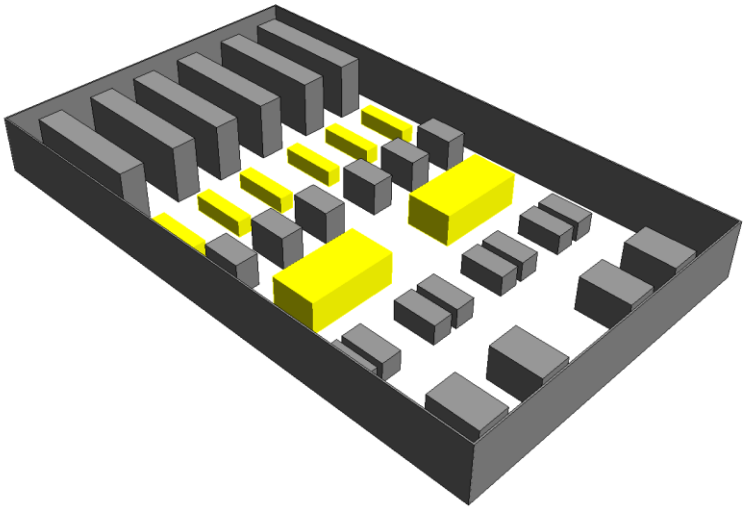


Figure 11. Highlighted elements based on utility parameters

If the user wants to create a separate drawing view for the material flow indicators, it is possible to use the Dynamo program to customize a new view. An example of a material flow visualization is shown in Figure 12.

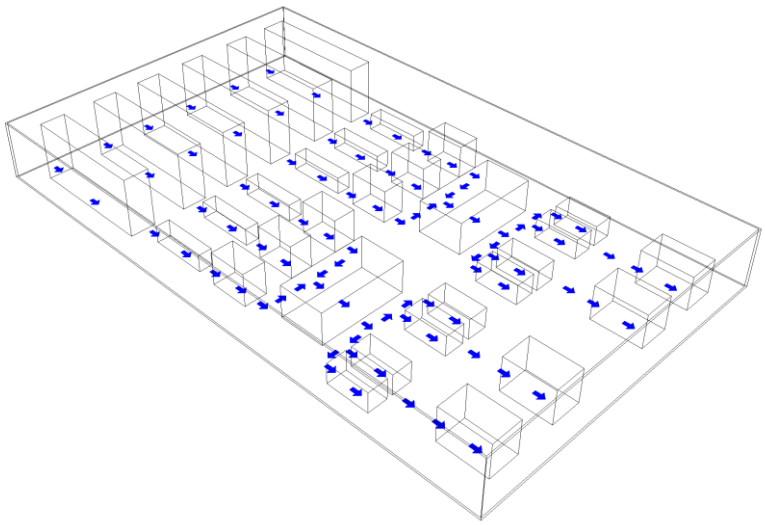


Figure 12. Material flow indicators within the factory

This tool can also be used to visualize separate views where people paths, AGV paths, or maintenance areas are highlighted. It can also be used to highlight HMIs or operators in the model. In Figure 13, the people flow paths inside the factory have been highlighted. If needed, specific elements can also be hidden when creating the view. This results in a fast and efficient way of sharing interface visualizations across the factory design team.

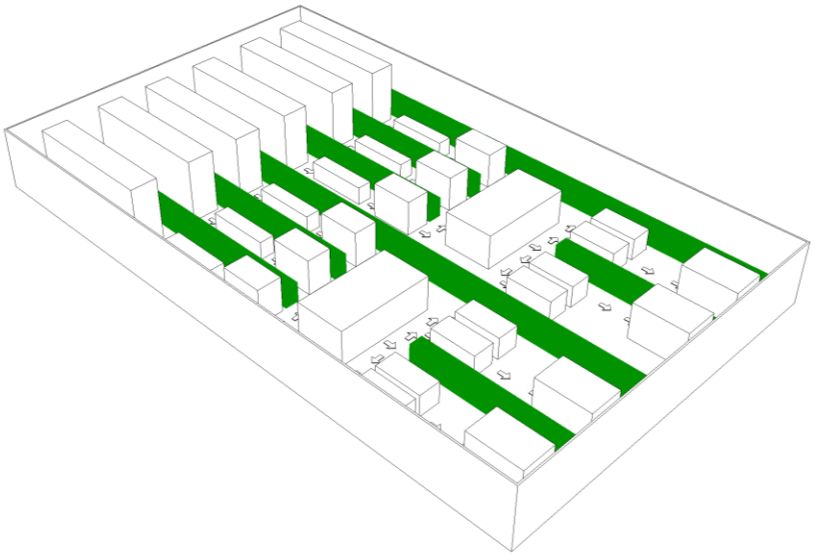


Figure 13. People flow paths inside the factory

The development of this Minimum Viable Product shows the potential of visualizing interfaces earlier and increases the overall understanding of the factory design. Even if the machine models have not reached LOD 350 and have all the required machine interfaces, the program can still visualize machines containing specific metadata, or show the preliminary placement of these interfaces. This gives the factory design disciplines a better overview of the factory design.

7 DISCUSSION

In the discussion chapter, the results and conclusions from the thesis are presented. Also, the literature and interviews will be discussed and compared with the developments which have been investigated.

7.1 Results

In general, this study provides a broader understanding of how to manage interfaces within the factory design process. A variety of studies mentioned in the literature review proves that interface management leads to better coordination and control between the different factory design disciplines. This is also clarified by the respondents in this study. Earlier interface visualization increases communication with equipment suppliers and enables better communication between the design disciplines.

The first step when developing an interface management workflow is to look at the current standards and company guidelines and give recommendations for how and when interfaces should be marked out. Companies within the production industry are heavily dependent on receiving the right information from the equipment suppliers. Understanding the total workspace area, also called the effective footprint, will be important according to both the literature studies and the interviews. The drawing template in Figure 5 has a separate drawing view where the surrounding interfaces, such as the material supply, operator area, and maintenance area can be marked out. The placement of these needs to be specified from the beginning. The utility interfaces can be placed in the models when a request for quotation (RFQ) is made, and the suppliers send detailed drawing information about their machines. This is required to happen within Basic design, and the models can then be developed to LOD 350. At detailed design, all interfaces and hook-up points will be assigned a reference tag. Further on, an ISO color standard can be used to highlight utility pipelines, interfaces, and hook-up points within the factory.

Besides standards and guidelines, this project has also investigated how interface drawing data can be visualized earlier in BIM. Figure 6 shows an example of how the machine could be visualized in a low maturity level. The machine is visualized as a generic placeholder but includes useful information regarding maintenance space, the material flow, and the operator placement. Since the maturity level is low, no utility interfaces will be shown. Information regarding utility consumption should be stored as metadata. Figure 7 shows an example of a machine at a medium maturity level. This machine shows more in detail how the machine looks like, and where the HMI panel will be located. Figure 8 shows an example of where all the surrounding interfaces are located and where the pipes should be fitted. Figure 11 shows how machines in a factory layout can be highlighted depending on the utilities they consume. This is possible even if the machine models have not reached LOD 350 since the machine element has attached metadata.

Figures 12 and 13 shows different views of how material flow indicators and people paths can be highlighted. These views create a better understanding of the elements within the factory, both for external people and for other design disciplines. A Minimum Viable Product (MVP) was created in Dynamo to show how separate interface elements can be visualized on a factory level. Dynamo

has a good user experience which enables fast development and testing. Another way to accomplish this is to create view templates in Revit where visibility/graphic overrides can be assigned. By using view templates, these can be directly loaded into a new factory project. Besides this, it is possible to create an add-in program for Revit. Another way to solve this could be to use Autodesk Platform Service (APS) and create a design automation add-in for Revit by converting the Dynamo program to a C# program. Autodesk Platform Service is a set of APIs and web servers which can be used to build cloud applications for models.

This study shows some examples of interfaces that can be highlighted. However, this study has been delimited to only investigate machine interfaces. The workflow could be further developed to show other interfaces and components in the factory such as rooms, doors, hatches, cable trays, fire sprinklers, or other things. However, this require that all model elements are categorized or have metadata attached so the model elements are searchable and can be filtered out.

7.2 Method

The foundation for this Master of Science thesis was the qualitative study with interviews conducted in different teams at the company. The qualitative study was a useful method of collecting data about the current process and ways of working. As described in the method chapter, all the respondents in the interviews have a technical background within engineering. While performing the interviews, some people recommended other people to be interviewed who had specific knowledge about specific topics. This was helpful in the process of finding the right people to interview. All interviews were conducted internally at the company, no external consultants or experts from academia have been interviewed, which can be seen as one delimitation of this study. To receive a more holistic perspective of the problem, it would have been good to interview external experts or consultants from other companies. These people could have shared their view of the situation and come up with alternative solutions for this could have been developed. However, due to sensitivity these people were not included in this process. Although, the results from this study do provide a good overview of the issues faced within the company and its future needs. In general, all respondents share the same perspective on the company and its internal resources.

Since the purpose of this thesis was to help the company to develop its interface management workflow, most of the relevant information has come from individual people at the company. There have been ongoing weekly discussions with the team and the factory layout manager. A lot of ideas for how to develop the workflow have come through these ongoing discussions. However, scientific literature validates the empirical findings and the need to manage interfaces. The literature also provides alternative solutions for how to detect, visualize, and solve interfacial issues. When transcribing and analyzing the interviews, much information could not be presented in this report due to confidentiality reasons. Based on that, a separate company report has been created which outlines the results and conclusions in detail. This refers mostly to the internal ways of working and the organization structures.

When all the interviews were performed and transcribed, the development stage of recommendations was initialized. The recommendations are based on the respondent’s answers

and their needs for visualizing more information in BIM. The work has been carried out in iterations, where developing testing and feedback has been ongoing simultaneously. The feedback received for each week, both from the team and outside of the team, has been helpful when continuing to develop the workflow.

7.3 The work in a wider content

The shift towards electrification is one of the largest transitions in modern history. To enable this, there is a need for sustainable Gigafactories to be built which can meet the increased demand for batteries and pave the way towards an electrified future. Since the transition towards electrification is happening at a rapid speed, these factories are built under tight deadlines. At the same time, factories should be as efficient, optimized, and reliable as possible. Batteries in electric vehicles are the most expensive element in an electric car today. The cost of building the Gigafactories will impact the price of each battery cell. This means that the designing, planning, and building process of Gigafactories will impact how much our electric vehicles and energy storage systems will cost in the future. As the literature review remarked, Interface management is potentially a way to increase efficiency through more streamlined communication and can decrease cost and planning time for new factories. However, there are a lot of other parameters which could have an impact on the final cost, for example, the material prices, the transportation prices, or the required number of operators within the factory. Lowering the factory's cost, without affecting battery performance or safety, will positively affect our lives both from an ethical and societal aspect.

Managing machine interfaces is just one aspect to consider when building large-scale factories. Another interesting topic for optimizing the factory layout would be to create a workflow for generative design in factory layouts. Generative design can help the layout engineers to produce alternative layouts that are more optimized. Optimization can be grounded on different parameters, for example minimizing material flow transportation, minimizing the size of the factory, or minimizing the length of the pipelines that distribute the machine with utilities. interfaces and total workspace areas will be needed when utilizing generative design programs. Generative design has been integrated into BIM in various studies already, for example by Abrishami et al. (2020). Based on the literature findings there are future challenges in implementing this. One of the stated problems by Wiendahl et al. (2015) is that generative design algorithms most of the time only consider one optimization criterion. However, this thesis topic lays the foundation for successfully implementing such workflow, since machine interfaces and the total workspace area will be needed. For developing a Generative design workflow, it is possible to start with Revit and Dynamo, since it already has capabilities for executing Generative design into the BIM models.

8 CONCLUSION

The conclusion chapter contains a summary of the purpose, research questions, and findings from the study. The chapter also mentions future studies within the field.

This study was initiated to investigate and further developing an efficient workflow for how to manage machine interfaces when designing large and complex production facilities. The project was delimited to investigate machine interfaces, which means all elements having direct contact with the machines in the factory. The following research questions are intended to be answered:

RQ1: What are the main challenges within factory modelling and interface management today?
RQ2: How can these interfaces be managed more effectively when designing Gigafactories?

The following challenges were identified in this study:

- Receiving standardized inputs with interface information from the equipment suppliers
- Understanding the total workspace area around the machine and how to visualize it
- The complexity of handling utility interfaces and visualize where these will be located

Designing a factory requires a lot of different design disciplines to work cross-functionally. Based on the study's findings, interface management can increase coordination and communication and at the same time decrease the cost of building factories. To manage interfaces more effectively, there needs to be a standardized process with clear definitions on how interfaces should be marked up and when in the design process they should be marked up. It is also important with standardized interface inputs from the equipment suppliers and to update the interface information throughout the design stages.

Further on, it is important to understand a machine's effective footprint in the factory, which is the total workspace area needed around the machine. The total workspace area includes for example the material flow and autonomous guided vehicle interface, the operator's workspace area and interaction with the machine, and the area needed for repair and maintenance. This study shows that utility interfaces for pipes, ducts, and electricity, are the most complex interfaces to handle. The study also reveals that interface markup will be done within different maturity levels. High maturity of interface information can only be visualized when the machine design is finalized. When the maturity level is low, it is hard to visualize the placement of the interfaces in the factory models. When a full system understanding of the factory is created and each machine has utility interfaces marked up, the hook-up points for utility distribution can be marked out. Sometimes, many machines can share the same hook-up point if their utility interfaces are located close to each other. This study also recommends when different interfaces can be visualized and updated in BIM.

The main goal of this research topic is to build more optimized and cost-efficient factories. Managing interfaces is just one step further to enable this. Future studies should investigate and measure the impact of these implementations. Future studies should also investigate generative design for factories and how this can be integrated. This can potentially create more optimized production lines and more efficient factories.

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APPENDIX A:1(2) INTERVIEW GUIDE - GENERAL

Respondent:
Role:
Date:

Summary:

I am doing my master thesis in the Factory Layout and Modelling team at Northvolt. The purpose of this project is to analyze the current 3D modelling workflow and see how we more effectively can manage the machine interfaces. All machines in the factory have different interfaces, for example, the physical attachment to the floor, utilities coming in and out, people paths, or Human-Machine Interfaces. A large focus of the project will be on how the 3D model can be optimized and become a central tool for effective communication in the initial phase of a new project and between all design disciplines.

- You will be anonymous throughout the report.
- Is it ok if I record this interview so I can go back and listen to it again? The recording will be deleted after the transcription of the interview is made.

Introduction questions:

1. Can you describe your job and what area you are working in?

2. Which design disciplines and people do you have the most contact with?
    a. Vendors? Contractors? Consultants?

Team-specific questions:

3. How often do you interact with the process layout (2D) and the BIM (3D)?
    a. Which programs are you interacting with?
    b. Do you see any disadvantages for how this is done today?

4. What is important for you to be displayed in the 2D or 3D model?
    a. Is something missing today?
    b. Should something be displayed earlier?

5. What comments do our consultants and vendors have on the process?
    a. Overall needs for future updates to the workflow?

Factory layout and Modelling questions:

6. The interfaces are currently not marked up in the process layout and BIM model from the start.
    a. Do you think it would be useful to have this information earlier?
    b. When is this information needed?
    c. Do you think it is needed to implement coordinates or connector points to specify the position of the utilities in the 3D model?

7. Some Architectural, Engineering, and Construction companies are fully utilizing the LOD (Level of Development) approach in their design, by having LOD 100, 200, 300, 350, 400, and 500 to ensure the right data is provided at the right time.
    a. Do you think it would be useful to fully implement LOD at Northvolt?
    b. Who would benefit? We or our stakeholders?

8. Do you have any other recommendations on implementations to be made?
    a. Specific for the 2D and 3D modelling workflow?
    b. Change in information flow?
    c. Changes to the design guidelines for how we get input from vendors?

Ending questions:

9. In the ideal case scenario for the future:
    a. How do you think your team should work in the upcoming years and how often should they work with 2D and 3D models?

Thank you for your time.

10. Do you have anything additional to say before we end this interview?

APPENDIX B:1(2) INTERVIEW GUIDE - LAYOUT

Respondent:
Role:
Date:

Summary:

I am doing my master thesis in the Factory Layout and Modelling team at Northvolt. The purpose of this project is to analyze the current 3D modelling workflow and see how we more effectively can manage the machine interfaces. All machines in the factory have different interfaces, for example, the physical attachment to the floor, utilities coming in and out, people paths, or Human-Machine Interfaces. A large focus of the project will be on how the 3D model can be optimized and become a central tool for effective communication in the initial phase of a new project and between all design disciplines.

- You will be anonymous throughout the report.
- Is it ok if I record this interview so I can go back and listen to it again? The recording will be deleted after the transcription of the interview is made.

Introduction questions:

1. Can you describe your job and what area you are working in?

2. Which design disciplines and people do you have the most contact with?
    a. Vendors? Contractors? Consultants?

Team-specific questions:

3. Can you explain more about the current workflow you are using?
    a. Which tools and processes are you using?
    b. What do you see as the most critical aspect of your current workflow?
    c. What is important for you to be displayed in the 2D or 3D model and when should it be displayed?

4. What are the most time-consuming tasks in your current way of working?
    a. Hook-up points?
    b. Inserting detailed 3D models?
    c. Clash coordination?
    d. MEP integration?
    e. Data transmittal between different design disciplines?

5. What comments do our consultants and vendors have on the process?
    a. Overall needs for future updates to the workflow?

Factory Layout Engineers – specific questions

6. The interfaces are currently not marked up in the process layout and BIM model from the start.
    a. Do you think it would be useful to implement this information earlier?
    b. Do you think coordinates or connector points are needed?

7. Some Architectural, Engineering, and Construction companies are fully utilizing the LOD (Level of Development) approach in their design, by having LOD 100, 200, 300, 350, 400, and 500 to ensure the right data is provided at the right time.
    a. Do you think it would be useful to fully implement LOD at Northvolt?
    b. Who would benefit? We or our stakeholders?

8. Do you have any other recommendations on implementations to be made?
    a. Specific for the 2D and 3D modelling workflow?
    b. Change in information flow?
    c. Changes to the design guidelines for how we get input from vendors?

Ending questions:

9. In the ideal case scenario for the future:
    a. How do you think your team should work in the upcoming years and to what extent should they work with 2D and 3D models?

Thank you for your time.

10. Do you have anything additional to say before we end this interview?

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